

## ECE 332 — Homework 3: Generated Answers from Final Cheat Sheet

Each solution shows the cheat sheet snippet used, then the answer.

**Symbol reference** (used throughout):

- $\epsilon_0 = 8.854 \times 10^{-12}$  F/m — permittivity of free space (constant)
- $\mu_0 = 4\pi \times 10^{-7}$  H/m — permeability of free space (constant)
- $\epsilon_r$  — relative permittivity (dielectric constant) of the medium, dimensionless
- $\mu_r$  — relative permeability of the medium, dimensionless.  $\mu_r = 1$  for nonmagnetic materials (air, most dielectrics)
- $\epsilon = \epsilon_r \epsilon_0$  — absolute permittivity of the medium (F/m)
- $\mu = \mu_r \mu_0$  — absolute permeability of the medium (H/m)
- $\sigma$  — conductivity of the medium (S/m).  $\sigma = 0$  for lossless / ideal dielectric
- $\eta = \sqrt{\mu/\epsilon}$  — intrinsic impedance of the medium ( $\Omega$ )
- $\eta_0 = \sqrt{\mu_0/\epsilon_0} = 120\pi \approx 377\Omega$  — intrinsic impedance of free space
- $k = \omega\sqrt{\mu\epsilon} = \omega\sqrt{\epsilon_r}/c$  (nonmag.) — wavenumber (rad/m)
- $\alpha$  — attenuation constant (Np/m);  $\beta$  — phase constant (rad/m)
- $\Gamma = (\eta_2 - \eta_1)/(\eta_2 + \eta_1)$  — reflection coefficient (unitless)
- $\tau = 1 + \Gamma$  — transmission coefficient (unitless)
- Superscripts on  $\tilde{\mathbf{E}}$ :  $i$  = incident,  $r$  = reflected,  $t$  = transmitted. Three distinct waves, three distinct superscripts.

***Note:** the official HW3 solution PDF labels both reflected and transmitted as  $\tilde{E}^r$  (typo). The second one is really  $\tilde{E}^t$  — you can tell because it carries  $\tau$  and  $k_2$  instead of  $\Gamma$  and  $k_1$ . This doc uses the correct  $i/r/t$  throughout.*

### Problem 1 — LHC Wave Normally Incident on a Dielectric

20 pts

**Setup:** 200 MHz LHC plane wave in air,  $|E| = 5$  V/m, normally incident on dielectric ( $\epsilon_r = 4$ ) at  $z \geq 0$ . Positive max at  $z = 0$ ,  $t = 0$ .

TABLE 8-2 —  $\Gamma, \tau, R, T$  SUMMARY

**Med. 1** ( $z < 0$ ) incident side; **Med. 2** ( $z \geq 0$ ) transmitted.  
 Shorthand:  $c_i = \cos \theta_i$ ,  $c_t = \cos \theta_t$ .

|                     | Normal<br>( $\theta_i = 0$ )              | $\perp$ pol                                               | $\parallel$ pol                                               |
|---------------------|-------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
| $\Gamma$ refl. wave | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$ | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$     |
| $\tau$ trans. wave  | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$             | $\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$                 |
| $\tau$ rel.         | $\tau = 1 + \Gamma$                       | $\tau_{\perp} = 1 + \Gamma_{\perp}$                       | $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{c_i}{c_t}$ |
| $R$ refl. pwr       | $ \Gamma ^2$                              | $ \Gamma_{\perp} ^2$                                      | $ \Gamma_{\parallel} ^2$                                      |
| $T$ trans. pwr      | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_{\perp} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          | $ \tau_{\parallel} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          |
| $R+T$ total         | $T = 1 - R$                               | $T_{\perp} = 1 - R_{\perp}$                               | $T_{\parallel} = 1 - R_{\parallel}$                           |

Notes:  $\sin \theta_t = \sqrt{\mu_1 \varepsilon_1 / \mu_2 \varepsilon_2} \sin \theta_i$ ; nonmag.  $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$ .

**Intrinsic impedance**  $\eta_i$  plug into table above

$$\eta_i = \sqrt{\frac{\mu_r}{\varepsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\varepsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default  $\mu_r = 1$  (air, dielectric). Use  $\mu_r \neq 1$  only if problem says "ferromagnetic"/iron/ferrite/etc.

**Low-loss correction**  $\eta_c \propto \sigma / (\omega \varepsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon}} \left( 1 + j \frac{\sigma}{2\omega \varepsilon} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\varepsilon_r}}$$

For  $\Gamma/\tau$  just use  $\eta_0 / \sqrt{\varepsilon_r}$ . The  $j$  term is for  $|\eta_c|, \theta_{\eta}$ . See LOSSY MEDIA — 5 CASES for all regimes.

### POWER REFLECTION & TRANSMISSION

% reflected *always*

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  *$\eta$ -ratio matters!*

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1/\eta_2$  factor.

### $\tilde{\mathbf{E}}$ PHASOR TEMPLATES (Air: $\varepsilon_r = \mu_r = 1$ )

Wavenumber per medium *nonmag.*

$$k_i = \frac{\omega \sqrt{\varepsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\varepsilon_{r,2}/\varepsilon_{r,1}}$$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$

| Inc dir    | Inc           | Refl          | Trans         |
|------------|---------------|---------------|---------------|
| $+\hat{u}$ | $e^{-jk_1 u}$ | $e^{+jk_1 u}$ | $e^{-jk_2 u}$ |
| $-\hat{u}$ | $e^{+jk_1 u}$ | $e^{-jk_1 u}$ | $e^{+jk_2 u}$ |

$+\hat{u}$  direction  $\Leftrightarrow$  Med 2 at  $u \geq 0$ . Reflected sign flips, transmitted matches incident.

General template (incident has pol.  $\hat{e}$ , amplitude  $a_0$ ):

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad (\text{incident wave, Med 1})$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad (\text{reflected wave, Med 1})$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace  $e^{\mp j k_2 u}$  with  $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$ )

Total in medium 1:  $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$ .

$\hat{e}$  by polarization *plug into templates above*

| Pol.               | $\hat{e}$                               | Note              |
|--------------------|-----------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                               | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                               | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y})/\sqrt{2}$          | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                    | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                    | $\delta = +\pi/2$ |
| General            | $a_x \hat{x} + a_y e^{j\delta} \hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

### Wave parameters in medium 1 (air):

Angular frequency from  $f$ :

$$\omega = 2\pi f = 2\pi(2 \times 10^8) = 4\pi \times 10^8 \text{ rad/s}$$

Wavenumber from the lossless plane wave formula  $k = \omega \sqrt{\varepsilon_r}/c$ . In air,  $\varepsilon_r = 1$  so this reduces to  $k_1 = \omega/c$ :

$$k_1 = \frac{\omega}{c} = \frac{4\pi \times 10^8}{3 \times 10^8} = \frac{4\pi}{3} \text{ rad/m}$$

(a) **LHC incident phasor.** The PHASOR TEMPLATES section says the incident wave has the form  $\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{-jk_1 z}$  where  $\hat{e}$  depends on polarization. From the polarization table:

$$\text{LHC} \Rightarrow \hat{e} = \hat{x} + j\hat{y}$$

With  $a_0 = 5 \text{ V/m}$  (the modulus) and  $k_1 = 4\pi/3$ :

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

**(b) Reflection coefficient  $\Gamma$  and transmission coefficient  $\tau$ .**

General intrinsic impedance:  $\eta = \sqrt{\mu/\varepsilon}$ .

Decompose into relative + free-space constants: since  $\mu = \mu_r \mu_0$  and  $\varepsilon = \varepsilon_r \varepsilon_0$ ,

$$\eta = \sqrt{\frac{\mu_r \mu_0}{\varepsilon_r \varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \cdot \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{\frac{\mu_r}{\varepsilon_r}} \eta_0$$

which for nonmagnetic media ( $\mu_r = 1$ ) reduces to  $\eta = \eta_0 / \sqrt{\varepsilon_r}$ .

Compute  $\eta_0$  from  $\mu_0, \varepsilon_0$  directly:

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = \sqrt{(4\pi \times 10^{-7})(36\pi \times 10^9)} = 120\pi \Omega \approx 377 \Omega$$

Medium 1 (air):  $\mu_r = 1, \varepsilon_r = 1$ . So  $\eta_1 = \eta_0 \sqrt{1/1} = \eta_0 = 120\pi \Omega$ .

Medium 2 (nonmagnetic dielectric,  $\mu_r = 1, \varepsilon_r = 4$ ): use the shortcut  $\eta = \eta_0 / \sqrt{\varepsilon_r}$ :

$$\eta_2 = \frac{\eta_0}{\sqrt{\varepsilon_{r2}}} = \frac{120\pi}{\sqrt{4}} = \frac{120\pi}{2} = 60\pi \Omega$$

Now apply the boundary formula  $\Gamma = (\eta_2 - \eta_1) / (\eta_2 + \eta_1)$ :

$$\Gamma = \frac{60\pi - 120\pi}{60\pi + 120\pi} = \frac{-60\pi}{180\pi} = \boxed{-\frac{1}{3}}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{3} = \boxed{\frac{2}{3}}$$

**(c) Reflected, transmitted, and total field phasors.**

Wavenumbers in both media use the same formula  $k_i = \omega \sqrt{\varepsilon_{r,i}} / c$ , just plug in each medium's  $\varepsilon_r$ :

Medium 1 (air,  $\varepsilon_{r,1} = 1$ ):

$$k_1 = \frac{\omega \sqrt{1}}{c} = \frac{4\pi \times 10^8}{3 \times 10^8} = \frac{4\pi}{3} \text{ rad/m}$$

Medium 2 (dielectric,  $\varepsilon_{r,2} = 4$ ):

$$k_2 = \frac{\omega \sqrt{4}}{c} = \frac{2\omega}{c} = 2k_1 = \frac{8\pi}{3} \text{ rad/m}$$

Notice: same formula, only  $\varepsilon_r$  changes per medium (just like  $\eta$ ).

From the PHASOR TEMPLATES (reflected wave travels in  $-\hat{z}$ , sign of  $z$ -term flips):

$$\tilde{\mathbf{E}}^r = \Gamma \cdot a_0 \hat{e} e^{+jk_1 z} = -\frac{1}{3} \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -\frac{5}{3}(\hat{x} + j\hat{y}) e^{+j4\pi z/3}$$

Transmitted wave in lossless medium 2:

$$\tilde{\mathbf{E}}^t = \tau \cdot a_0 \hat{e} e^{-jk_2 z} = \frac{2}{3} \cdot 5(\hat{x} + j\hat{y}) e^{-j8\pi z/3} = \frac{10}{3}(\hat{x} + j\hat{y}) e^{-j8\pi z/3}$$

Total field in medium 1 is incident + reflected:

$$\boxed{\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x} + j\hat{y}) \left( e^{-j4\pi z/3} - \frac{1}{3} e^{+j4\pi z/3} \right) \text{ V/m}}$$

**(d) Percentages of incident average power reflected and transmitted.**

From the *POWER REFLECTION* snippet,  $\%P_r = 100|\Gamma|^2$  and  $\%P_t = 100|\tau|^2(\eta_1/\eta_2)$ . The  $\eta_1/\eta_2$  factor accounts for the change in wave impedance: even though  $|\tau| > 0$ , the medium beyond carries power differently.

$$\%P_r = 100 \cdot \left| -\frac{1}{3} \right|^2 = 100 \cdot \frac{1}{9} = \boxed{11.1\%}$$

$$\%P_t = 100 \cdot \left| \frac{2}{3} \right|^2 \cdot \frac{120\pi}{60\pi} = 100 \cdot \frac{4}{9} \cdot 2 = \boxed{88.9\%}$$

*Sanity check:*  $11.1\% + 88.9\% = 100\%$ . Energy conserved.

## Problem 2 — Same Wave, Poor-Conductor Medium 2

20 pts

**Setup:** Replace medium 2 with poor conductor:  $\epsilon_r = 2.25$ ,  $\mu_r = 1$ ,  $\sigma = 10^{-4}$  S/m. Everything else same as P1.

TABLE 8-2 —  $\Gamma, \tau, R, T$  SUMMARY

**Med. 1** ( $z < 0$ ) incident side; **Med. 2** ( $z \geq 0$ ) transmitted.  
Shorthand:  $c_i = \cos \theta_i$ ,  $c_t = \cos \theta_t$ .

|                     | Normal<br>( $\theta_i = 0$ )              | $\perp$ pol                                               | $\parallel$ pol                                               |
|---------------------|-------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
| $\Gamma$ refl. wave | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$ | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$     |
| $\tau$ trans. wave  | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$             | $\frac{2\eta_2 c_t}{\eta_2 c_t + \eta_1 c_i}$                 |
| $\tau$ rel.         | $\tau = 1 + \Gamma$                       | $\tau_{\perp} = 1 + \Gamma_{\perp}$                       | $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{c_i}{c_t}$ |
| $R$ refl. pwr       | $ \Gamma ^2$                              | $ \Gamma_{\perp} ^2$                                      | $ \Gamma_{\parallel} ^2$                                      |
| $T$ trans. pwr      | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_{\perp} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          | $ \tau_{\parallel} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          |
| $R+T$ total         | $T = 1 - R$                               | $T_{\perp} = 1 - R_{\perp}$                               | $T_{\parallel} = 1 - R_{\parallel}$                           |

Notes:  $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$ ; nonmag.  $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$ .

Intrinsic impedance  $\eta_i$  plug into table above

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} \quad (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

Default  $\mu_r = 1$  (air, dielectric). Use  $\mu_r \neq 1$  only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction  $\eta_c \approx \sigma / (\omega \epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon}} \left( 1 + j \frac{\sigma}{2\omega \epsilon} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

For  $\Gamma/\tau$  just use  $\eta_0 / \sqrt{\epsilon_r}$ . The  $j$  term is for  $|\eta_c|, \theta_{\eta}$ . See LOSSY MEDIA — 5 CASES for all regimes.

## LOSSY MEDIA — 5 CASES

Loss tangent  $\epsilon' = \epsilon_r \epsilon_0$ ,  $\epsilon_0 \approx 1 / (36\pi \times 10^9)$

$$\frac{\epsilon''}{\epsilon'} = \frac{\sigma}{\omega \epsilon_r \epsilon_0}$$

| Type                         | $\sigma / \omega \epsilon$ | $\eta_c, \alpha, \beta$                                                                                                                                                                                               |
|------------------------------|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lossless<br>( $\sigma = 0$ ) | 0                          | $\eta = \eta_0 / \sqrt{\epsilon_r}$ exact<br>$\alpha = 0, \beta = \omega \sqrt{\epsilon_r} / c$                                                                                                                       |
| Low-loss                     | $< 10^{-2}$                | $\eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \left( 1 + j \frac{\sigma}{2\omega \epsilon} \right)$<br>$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}}, \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c}$ |
| Quasi-cond.                  | $10^{-2} - 10^2$           | exact: $(\eta / \sqrt{X}) e^{j\theta_{\eta}}$ , see below                                                                                                                                                             |
| Good cond.                   | $> 10^2$                   | $\eta_c \approx (1+j) \sqrt{\pi f \mu / \sigma}$<br>$\alpha \approx \beta \approx \sqrt{\pi f \mu \sigma}$                                                                                                            |
| Magnetic                     | any                        | keep full $\sqrt{\mu_r / \epsilon_r} \eta_0$                                                                                                                                                                          |

For  $\Gamma/\tau$ : drop  $j$  correction; use  $\eta_0 / \sqrt{\epsilon_r}$ . Magnetic: keep  $\mu_r$ .

Low-loss quick params  $\sigma / (\omega \epsilon) \ll 1$ ,  $\mu_r = 1$

$$\alpha \approx \frac{\sigma \eta_0}{2\sqrt{\epsilon_r}} \text{ (Np/m)}, \quad \beta \approx \frac{\omega \sqrt{\epsilon_r}}{c} \text{ (rad/m)}, \quad \eta_c \approx \frac{\eta_0}{\sqrt{\epsilon_r}} \text{ (}\Omega\text{)}$$

Exact (quasi-cond)  $X = \sqrt{1 + (\epsilon'' / \epsilon')^2}$

$$\alpha = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X - 1}$$

$$\beta = \omega \sqrt{\frac{\mu \epsilon'}{2}} \sqrt{X + 1}$$

$$\theta_{\eta} = \frac{1}{2} \arctan \frac{\sigma}{\omega \epsilon'}$$

 $\tilde{\mathbf{E}}$  PHASOR TEMPLATES (Air:  $\epsilon_r = \mu_r = 1$ )

Wavenumber per medium nonmag.

$$k_i = \frac{\omega \sqrt{\epsilon_{r,i}}}{c} \text{ (rad/m)} \Rightarrow k_2 = k_1 \sqrt{\epsilon_{r,2} / \epsilon_{r,1}}$$

Direction table  $u \in \{x, y, z\} = \text{axis perp. to boundary}$

| Inc dir    | Inc           | Refl          | Trans         |
|------------|---------------|---------------|---------------|
| $+\hat{u}$ | $e^{-jk_1 u}$ | $e^{+jk_1 u}$ | $e^{-jk_2 u}$ |
| $-\hat{u}$ | $e^{+jk_1 u}$ | $e^{-jk_1 u}$ | $e^{+jk_2 u}$ |

$+\hat{u}$  direction  $\Leftrightarrow$  Med 2 at  $u \geq 0$ . Reflected sign flips, transmitted matches incident.

General template (incident has pol.  $\hat{e}$ , amplitude  $a_0$ ):

$$\tilde{\mathbf{E}}^i = a_0 \hat{e} e^{\mp j k_1 u} \text{ (V/m)} \quad (\text{incident wave, Med 1})$$

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{\pm j k_1 u} \text{ (V/m)} \quad (\text{reflected wave, Med 1})$$

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{\mp j k_2 u} \text{ (V/m)} \quad (\text{transmitted wave, Med 2})$$

(lossy med 2: replace  $e^{\mp j k_2 u}$  with  $e^{\mp \alpha_2 u} e^{\mp j \beta_2 u}$ )

Total in medium 1:  $\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r$ .

$\hat{e}$  by polarization plug into templates above

| Pol.               | $\hat{e}$                               | Note              |
|--------------------|-----------------------------------------|-------------------|
| Lin. $\hat{x}$     | $\hat{x}$                               | $E_y = 0$         |
| Lin. $\hat{y}$     | $\hat{y}$                               | $E_x = 0$         |
| Lin. at $45^\circ$ | $(\hat{x} + \hat{y}) / \sqrt{2}$        | in phase          |
| RHC                | $\hat{x} - j\hat{y}$                    | $\delta = -\pi/2$ |
| LHC                | $\hat{x} + j\hat{y}$                    | $\delta = +\pi/2$ |
| General            | $a_x \hat{x} + a_y e^{j\delta} \hat{y}$ | elliptical        |

$\Gamma, \tau$  act on  $\hat{e}$  unchanged. Only swap the polarization, not the formulas.

## POWER REFLECTION &amp; TRANSMISSION

% reflected always

$$\%P_r = 100 |\Gamma|^2$$

% transmitted  $\eta$ -ratio matters!

$$\%P_t = 100 |\tau|^2 \frac{\eta_1}{\eta_2}$$

Check:  $\%P_r + \%P_t = 100$ .

If medium 2 is denser ( $\eta_2 < \eta_1$ ),  $|\tau| > 1$  but  $\%P_t < 100\%$  because of the  $\eta_1 / \eta_2$  factor.

**Wave parameters in medium 1 (air).** Same as P1, since medium 1 hasn't changed:

$$\omega = 2\pi f = 4\pi \times 10^8 \text{ rad/s}, \quad k_1 = \frac{\omega \sqrt{\epsilon_{r,1}}}{c} = \frac{\omega}{c} = \frac{4\pi}{3} \text{ rad/m}, \quad \eta_1 = \eta_0 = 120\pi \Omega$$

**(a) LHC incident phasor.** Identical to P1 — incident wave is in medium 1 (air), which didn't change. From the PHASOR TEMPLATES + polarization table ( $\hat{e}_{\text{LHC}} = \hat{x} + j\hat{y}$ ):

$$\tilde{\mathbf{E}}^i = 5(\hat{x} + j\hat{y}) e^{-j4\pi z/3} \text{ V/m}$$

**(b) Reflection and transmission coefficients — but first classify medium 2.**

Compute the loss tangent  $\sigma/(\omega\epsilon)$  to pick which formulas to use:

$$\frac{\sigma_2}{\omega\epsilon_2} = \frac{\sigma_2}{\omega\epsilon_{r,2}\epsilon_0} = \frac{10^{-4}}{(4\pi \times 10^8)(2.25)(8.854 \times 10^{-12})} = 4 \times 10^{-3}$$

This is  $\ll 1$ , so medium 2 is a **low-loss dielectric**. The LOSSY MEDIA — 5 CASES table classifies media by  $\sigma/(\omega\epsilon)$  and tells us to use the *low-loss row* shortcuts (not the full exact lossy formulas in the bottom).

*Intrinsic impedance — same derivation chain as P1, with the low-loss correction.* The general formula  $\eta = \sqrt{\mu_r/\epsilon_r} \eta_0$  lives in the intrinsic-impedance footer of TABLE 8-2. For low-loss media the low-loss row of LOSSY MEDIA — 5 CASES says:

$$\eta_2 \approx \sqrt{\frac{\mu}{\epsilon'}} (1 + j \frac{\sigma}{2\omega\epsilon'}) \xrightarrow[\sigma/\omega\epsilon \ll 1]{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

The  $j$  correction is  $4 \times 10^{-3}/2 = 2 \times 10^{-3}$ , tiny enough to drop for  $\Gamma, \tau$  computation. So:

$$\eta_2 \approx \frac{\eta_0}{\sqrt{\epsilon_{r,2}}} = \frac{120\pi}{\sqrt{2.25}} = \frac{120\pi}{1.5} = 80\pi \Omega$$

*Attenuation and phase constants from low-loss row of LOSSY MEDIA — 5 CASES* (these matter for the transmitted-wave decay even though they don't affect  $\Gamma$ ):

$$\alpha_2 \approx \frac{\sigma_2 \eta_0}{2\sqrt{\epsilon_{r,2}}} = \frac{10^{-4} \cdot 120\pi}{2\sqrt{2.25}} = 1.26 \times 10^{-2} \text{ Np/m}$$

$$\beta_2 \approx \frac{\omega \sqrt{\epsilon_{r,2}}}{c} = \frac{(4\pi \times 10^8)(1.5)}{3 \times 10^8} = 2\pi \text{ rad/m}$$

Apply the boundary formulas:

$$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{80\pi - 120\pi}{80\pi + 120\pi} = \frac{-40\pi}{200\pi} = \boxed{-\frac{1}{5}}$$

$$\tau = 1 + \Gamma = 1 - \frac{1}{5} = \boxed{\frac{4}{5}}$$

**(c) Reflected, transmitted, and total field phasors.**

*Direction signs from the DIRECTION TABLE.* Med 2 is at  $z \geq 0$  (same as P1), so use the first row: incident gets  $-jk_1 z$ , reflected gets  $+jk_1 z$ , transmitted gets the lossy decay  $e^{-\alpha_2 z} e^{-j\beta_2 z}$  (since med 2 is now lossy, replace  $e^{-jk_2 z}$  with  $e^{-\alpha_2 z} e^{-j\beta_2 z}$ ).

*Reflected template (still in lossless medium 1, traveling  $-\hat{z}$ ):*

$$\tilde{\mathbf{E}}^r = \Gamma a_0 \hat{e} e^{+jk_1 z}$$

Plug in  $\Gamma = -1/5$ ,  $a_0 = 5$ ,  $\hat{e} = \hat{x} + j\hat{y}$ ,  $k_1 = 4\pi/3$ :

$$\tilde{\mathbf{E}}^r = -\frac{1}{5} \cdot 5(\hat{x} + j\hat{y}) e^{+j4\pi z/3} = -(\hat{x} + j\hat{y}) e^{+j4\pi z/3} \text{ V/m}$$

Transmitted template (lossy medium 2, traveling  $+\hat{z}$ , replace  $e^{-jk_2 z}$  with  $e^{-\alpha_2 z} e^{-j\beta_2 z}$ ):

$$\tilde{\mathbf{E}}^t = \tau a_0 \hat{e} e^{-\alpha_2 z} e^{-j\beta_2 z}$$

Plug in  $\tau = 4/5$ ,  $a_0 = 5$ ,  $\hat{e} = \hat{x} + j\hat{y}$ ,  $\alpha_2 = 1.26 \times 10^{-2}$ ,  $\beta_2 = 2\pi$ :

$$\tilde{\mathbf{E}}^t = \frac{4}{5} \cdot 5(\hat{x} + j\hat{y}) e^{-\alpha_2 z} e^{-j\beta_2 z} = 4(\hat{x} + j\hat{y}) e^{-1.26 \times 10^{-2} z} e^{-j2\pi z} \text{ V/m}$$

Total field in medium 1 ( $z \leq 0$ ):

$$\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r = 5(\hat{x} + j\hat{y}) \left( e^{-j4\pi z/3} - \frac{1}{5} e^{+j4\pi z/3} \right) \text{ V/m}$$

**(d) Percentages of incident average power reflected and transmitted.**

From *POWER REFLECTION* snippet, same formulas as P1:

$$\%P_r = 100|\Gamma|^2 = 100 \cdot \left| -\frac{1}{5} \right|^2 = 100 \cdot \frac{1}{25} = \boxed{4\%}$$

$$\%P_t = 100|\tau|^2 \frac{\eta_1}{\eta_2} = 100 \cdot \left| \frac{4}{5} \right|^2 \cdot \frac{120\pi}{80\pi} = 100 \cdot \frac{16}{25} \cdot \frac{3}{2} = \boxed{96\%}$$

*Sanity check:*  $4\% + 96\% = 100\%$ . Less reflection than P1 (4% vs 11.1%) because medium 2 is closer in impedance to air ( $\eta_2 = 80\pi$  vs  $60\pi$ ); the smaller impedance jump means more of the wave gets through.

**Problem 3 — Parallel-Polarized Wave Through Two Dielectric Layers 10 pts**

**Setup:**  $\theta_i = 30^\circ$  from air, layer 1 ( $\epsilon_r = 6.25$ ,  $t = 5$  cm), layer 2 ( $\epsilon_r = 2.25$ ,  $t = 5$  cm), exits to air.

**SNELL'S LAW — OBLIQUE INCIDENCE**

**Single boundary** *angles from normal*

$$\sin \theta_2 = \sin \theta_1 \sqrt{\frac{\epsilon_{r1}}{\epsilon_{r2}}} = \sin \theta_1 \frac{n_1}{n_2}$$

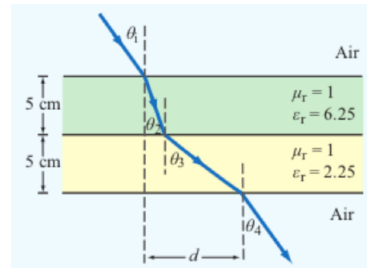
**Multilayer chain** *stack of slabs 1  $\rightarrow$  2  $\rightarrow$  3  $\rightarrow$  4*

$$\sin \theta_{i+1} = \sin \theta_i \sqrt{\frac{\epsilon_{r,i}}{\epsilon_{r,i+1}}}$$

**Air–slabs–air shortcut** *outer media identical*

$$\theta_{\text{exit}} = \theta_{\text{incident}}$$

*Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).*

**LATERAL DISPLACEMENT**

**Beam offset through  $N$  slabs** *thickness  $t_i$ , refracted angle inside slab  $i$*

**Slab-indexed**  $\theta_i =$  angle inside slab  $i$

$$d = \sum_{i=1}^N t_i \tan \theta_i$$

**Angle-indexed (HW3 P3 style)**  $\theta_1 =$  incident,  $\theta_{i+1} =$  inside slab  $i$

$$d = \sum_{i=1}^N t_i \tan \theta_{i+1}$$

*Same physics, different numbering. In HW3 P3: slab 1 contributes  $t_1 \tan \theta_2$ , slab 2 contributes  $t_2 \tan \theta_3$ . The angle in the formula is always the one the ray makes inside that slab.*

*Air–slab–air: exit angle returns to  $\theta_1$  but is laterally shifted by  $d$ .*



**TABLE 8-2 —  $\Gamma, \tau, R, T$  SUMMARY**

*Med. 1 ( $z < 0$ ) incident side; Med. 2 ( $z \geq 0$ ) transmitted.  
Shorthand:  $c_i = \cos \theta_i$ ,  $c_t = \cos \theta_t$ .*

|                     | Normal<br>( $\theta_i = 0$ )              | $\perp$ pol                                               | $\parallel$ pol                                               |
|---------------------|-------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------|
| $\Gamma$ refl. wave | $\frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$ | $\frac{\eta_2 c_i - \eta_1 c_t}{\eta_2 c_i + \eta_1 c_t}$ | $\frac{\eta_2 c_t - \eta_1 c_i}{\eta_2 c_t + \eta_1 c_i}$     |
| $\tau$ trans. wave  | $\frac{2\eta_2}{\eta_2 + \eta_1}$         | $\frac{2\eta_2 c_i}{\eta_2 c_i + \eta_1 c_t}$             | $\frac{2\eta_2 c_i}{\eta_2 c_t + \eta_1 c_i}$                 |
| $\tau$ rel.         | $\tau = 1 + \Gamma$                       | $\tau_{\perp} = 1 + \Gamma_{\perp}$                       | $\tau_{\parallel} = (1 + \Gamma_{\parallel}) \frac{c_i}{c_t}$ |
| $R$ refl. pwr       | $ \Gamma ^2$                              | $ \Gamma_{\perp} ^2$                                      | $ \Gamma_{\parallel} ^2$                                      |
| $T$ trans. pwr      | $ \tau ^2 \frac{\eta_1}{\eta_2}$          | $ \tau_{\perp} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          | $ \tau_{\parallel} ^2 \frac{\eta_1 c_t}{\eta_2 c_i}$          |
| $R + T$ total       | $T = 1 - R$                               | $T_{\perp} = 1 - R_{\perp}$                               | $T_{\parallel} = 1 - R_{\parallel}$                           |

*Notes:  $\sin \theta_t = \sqrt{\mu_1 \epsilon_1 / \mu_2 \epsilon_2} \sin \theta_i$ ; nonmag.  $\Rightarrow \eta_2 / \eta_1 = n_1 / n_2$ .*

**Intrinsic impedance  $\eta_i$**  *plug into table above*

$$\eta_i = \sqrt{\frac{\mu_r}{\epsilon_r}} \eta_0 \xrightarrow{\mu_r=1} \frac{\eta_0}{\sqrt{\epsilon_{r,i}}} (\Omega) \quad \eta_0 = 120\pi \approx 377 \Omega$$

*Default  $\mu_r = 1$  (air, dielectric). Use  $\mu_r \neq 1$  only if problem says “ferromagnetic”/iron/ferrite/etc.*

**Low-loss correction  $\eta_c$**   $\sigma / (\omega \epsilon) \ll 1$

$$\eta_c \approx \sqrt{\frac{\mu}{\epsilon'}} \left( 1 + j \frac{\sigma}{2\omega \epsilon'} \right) \xrightarrow{\text{drop } j} \frac{\eta_0}{\sqrt{\epsilon_r}}$$

*For  $\Gamma/\tau$  just use  $\eta_0 / \sqrt{\epsilon_r}$ . The  $j$  term is for  $|\eta_c|, \theta_{\eta}$ . See **LOSSY MEDIA — 5 CASES** for all regimes.*

(a) **Chained Snell’s law:**

$$\sin \theta_2 = \sin 30^\circ \sqrt{\frac{1}{6.25}} = 0.5 \cdot 0.4 = 0.2 \Rightarrow \theta_2 = \sin^{-1}(0.2) = \boxed{11.54^\circ}$$

$$\sin \theta_3 = 0.2 \sqrt{\frac{6.25}{2.25}} = 0.2 \cdot \frac{5}{3} = \frac{1}{3} \Rightarrow \theta_3 = \sin^{-1}(1/3) = \boxed{19.47^\circ}$$

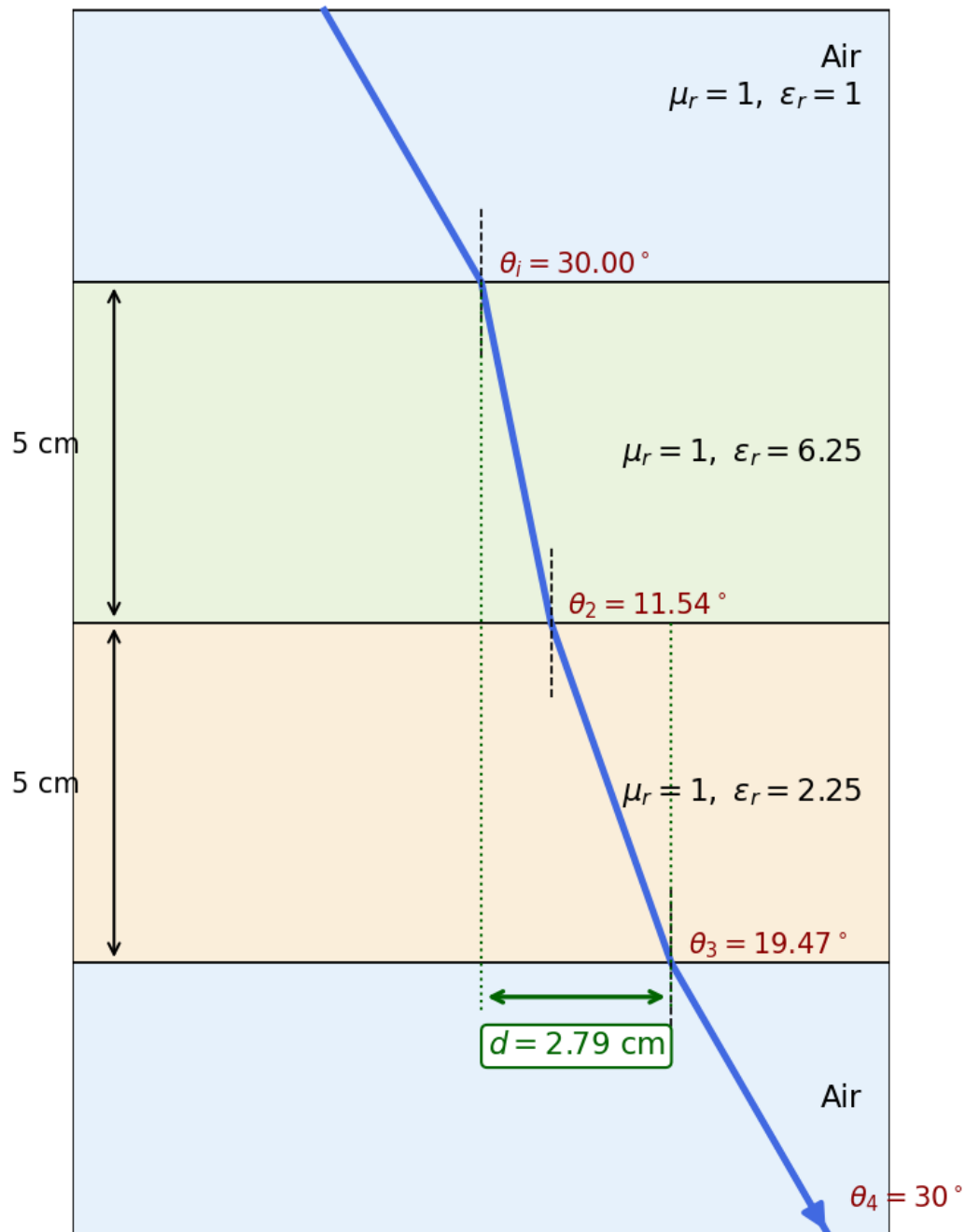
$$\sin \theta_4 = \frac{1}{3} \sqrt{\frac{2.25}{1}} = \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2} \Rightarrow \theta_4 = \boxed{30^\circ = \theta_i}$$

(b) **Lateral displacement:**

$$d = t_1 \tan \theta_2 + t_2 \tan \theta_3 = 5 \tan 11.54^\circ + 5 \tan 19.47^\circ = 5(0.2041 + 0.3536) \text{ cm}$$

$$\boxed{d = 2.79 \text{ cm}}$$

## HW3 P3: parallel-polarized wave through dielectric layers



## Problem 4 — Dielectric Waveguide TE Mode Identification

20 pts

**Setup:** TE wave,  $a = 5$  cm,  $b = 3$  cm,  $E_x = -36 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z)$  V/m.

## RECTANGULAR WAVEGUIDE — MODES

TE mode  $E_z = 0, H_z \neq 0$ TM mode  $H_z = 0, E_z \neq 0$ Dimensions  $a \times b$  with  $a > b$  along  $\hat{x}, \hat{y}$ ; wave in  $\hat{z}$ . $E_x$  form (TE) read  $m, n$  from arguments

$$E_x \propto \cos\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \sin(\omega t - \beta z)$$

coef. on  $x = m\pi/a$ ; coef. on  $y = n\pi/b$ .Identify TE <sub>$m$  $n$</sub>  vs TM <sub>$m$  $n$</sub>  cos/sin pattern of  $E_x$  is identical for both!

1. Read the problem statement (e.g. “a TE wave” → TE).
2. If  $m = 0$  or  $n = 0$  in the mode label  $\Rightarrow$  must be TE (TM forbids 0).
3. Source field  $H_z(x, y)$  given  $\Rightarrow$  TE. Source field  $E_z(x, y)$  given  $\Rightarrow$  TM.

TE allows at least one of  $m, n \geq 1$  (other can be 0). TM needs both  $m, n \geq 1$ . That's why TE<sub>10</sub> exists but TM<sub>10</sub> doesn't.

 $\epsilon_r$  FROM DISPERSIONGiven  $\omega, \beta$ , mode  $(m, n)$  result is unitless

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] \text{ (unitless)}$$

Inputs:  $\omega$  (rad/s),  $\beta$  (rad/m),  $a, b$  (m),  $c = 3 \times 10^8$  (m/s),  $m, n$  (unitless).

WAVE IMPEDANCE &  $H_y$ 

$$Z_{TE} = \frac{\eta}{\sqrt{1 - (f_c/f)^2}} \text{ (}\Omega\text{)} \quad Z_{TM} = \eta \sqrt{1 - (f_c/f)^2}$$

$$\eta = \frac{\eta_0}{\sqrt{\epsilon_r}} \quad H_y = \frac{E_x}{Z_{TE}} \text{ (TE mode)}$$

$Z_{TE} > \eta$  always (denominator  $< 1$ );  $Z_{TM} < \eta$ .

## TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand:  $C_X = \cos \frac{m\pi x}{a}$ ,  $S_X = \sin \frac{m\pi x}{a}$ ,  $C_Y = \cos \frac{n\pi y}{b}$ ,  $S_Y = \sin \frac{n\pi y}{b}$ . All fields carry an  $e^{-j\beta z}$  factor (omitted below).  $k_c^2 = (m\pi/a)^2 + (n\pi/b)^2$ .

TE mode generator:  $\tilde{H}_z$ 

$$\tilde{H}_z = H_0 C_X C_Y$$

$$\tilde{E}_x = \frac{j\omega\mu}{k_c^2} \left(\frac{n\pi}{b}\right) H_0 C_X S_Y$$

$$\tilde{E}_y = -\frac{j\omega\mu}{k_c^2} \left(\frac{m\pi}{a}\right) H_0 S_X C_Y$$

$$\tilde{E}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TE}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TE}$$

TM mode generator:  $\tilde{E}_z$ 

$$\tilde{E}_z = E_0 S_X S_Y$$

$$\tilde{E}_x = -\frac{j\beta}{k_c^2} \left(\frac{m\pi}{a}\right) E_0 C_X S_Y$$

$$\tilde{E}_y = -\frac{j\beta}{k_c^2} \left(\frac{n\pi}{b}\right) E_0 S_X C_Y$$

$$\tilde{H}_z = 0, \quad \tilde{H}_x = -\tilde{E}_y/Z_{TM}, \quad \tilde{H}_y = \tilde{E}_x/Z_{TM}$$

TEM (plane wave) for reference:  $\tilde{E}_x = E_0 e^{-j\beta z}$ ,  $\tilde{H}_y = \tilde{E}_x/\eta$ ,  $f_c$  N/A,  $k = \omega\sqrt{\mu\epsilon}$ .

## CUTOFF FREQUENCY

 $f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs ( $a > b$ )

| Quantity                            | Formula                                       |
|-------------------------------------|-----------------------------------------------|
| $u_{p0}$                            | $c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$ ) |
| TE <sub>10</sub>                    | $u_{p0}/(2a)$ (dominant)                      |
| TE <sub>01</sub>                    | $u_{p0}/(2b)$                                 |
| TE <sub>20</sub>                    | $u_{p0}/a = 2f_{10}$                          |
| TE <sub>11</sub> , TM <sub>11</sub> | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$       |

Propagation requires  $f > f_{mn}$ .

(a) **Mode from the field expression.** Read  $m\pi/a$  from the cos coefficient and  $n\pi/b$  from the sin coefficient.

$$\frac{m\pi}{a} = 40\pi \Rightarrow m = 40 \cdot 0.05 = 2$$

$$\frac{n\pi}{b} = 100\pi \Rightarrow n = 100 \cdot 0.03 = 3$$

Mode: TE<sub>23</sub>

(b)  $\epsilon_r$  from dispersion, with  $\omega = 2.4\pi \times 10^{10}$  rad/s,  $\beta = 52.9\pi$  rad/m:

$$\epsilon_r = \frac{c^2}{\omega^2} \left[ \beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2 \right] = \left( \frac{3 \times 10^8}{2.4\pi \times 10^{10}} \right)^2 [(52.9\pi)^2 + (40\pi)^2 + (100\pi)^2]$$

$$\epsilon_r = 2.25$$

(c) Cutoff frequency for  $\text{TE}_{23}$ :

$$u_{p0} = \frac{c}{\sqrt{\epsilon_r}} = \frac{3 \times 10^8}{1.5} = 2 \times 10^8 \text{ m/s}$$

$$f_{23} = \frac{u_{p0}}{2} \sqrt{\left(\frac{2}{a}\right)^2 + \left(\frac{3}{b}\right)^2} = 10^8 \sqrt{(2/0.05)^2 + (3/0.03)^2} = 10^8 \sqrt{1600 + 10000}$$

$$\boxed{f_{23} = 10.77 \text{ GHz}}$$

(d)  $H_y$  from  $Z_{\text{TE}}$ : operating frequency  $f = \omega/(2\pi) = 12 \text{ GHz}$ .

$$Z_{\text{TE}} = \frac{\eta}{\sqrt{1-(f_c/f)^2}} = \frac{377/\sqrt{2.25}}{\sqrt{1-(10.77/12)^2}} = \frac{251.3}{\sqrt{1-0.805}} = \frac{251.3}{0.442} = 569.9 \text{ } \Omega$$

$$\boxed{H_y = \frac{E_x}{Z_{\text{TE}}} = -0.063 \cos(40\pi x) \sin(100\pi y) \sin((2.4\pi \times 10^{10})t - 52.9\pi z) \text{ A/m}}$$

**Problem 5 — Hollow Guide, Multi-Mode Excitation & Travel Times** 8 pts

**Setup:** Hollow guide ( $\epsilon_r = 1$ ,  $u_{p0} = c$ ),  $a = 3$  cm,  $b = 2$  cm. Pulse with 9.5 GHz carrier, guide length 100 m.

**CUTOFF FREQUENCY**

$f_{mn}$   $m, n$  integers  $\geq 0$  (not both zero)

$$f_{mn} = \frac{u_{p0}}{2} \sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2} \text{ (Hz)}$$

Common cutoffs ( $a > b$ )

| Quantity                            | Formula                                       |
|-------------------------------------|-----------------------------------------------|
| $u_{p0}$                            | $c/\sqrt{\epsilon_r}$ (hollow: $u_{p0} = c$ ) |
| TE <sub>10</sub>                    | $u_{p0}/(2a)$ (dominant)                      |
| TE <sub>01</sub>                    | $u_{p0}/(2b)$                                 |
| TE <sub>20</sub>                    | $u_{p0}/a = 2f_{10}$                          |
| TE <sub>11</sub> , TM <sub>11</sub> | $\frac{u_{p0}}{2} \sqrt{1/a^2 + 1/b^2}$       |

Propagation requires  $f > f_{mn}$ .

**GROUP VELOCITY & TRAVEL TIME**

$$u_g = u_{p0} \sqrt{1 - (f_{mn}/f)^2} \text{ (m/s)}$$

$$u_p = \frac{u_{p0}}{\sqrt{1 - (f_{mn}/f)^2}} \text{ (m/s)} \Rightarrow u_p u_g = u_{p0}^2 \text{ (m}^2\text{/s}^2\text{)}$$

$t = L/u_g$  (s) travel time through guide of length  $L$   
Higher modes have lower  $u_g \Rightarrow$  they arrive later. Group delay spread = sum over excited modes.

Cutoff frequencies for the lowest modes:

$$f_{10} = \frac{c}{2a} = \frac{3 \times 10^8}{0.06} = 5 \text{ GHz}$$

$$f_{01} = \frac{c}{2b} = \frac{3 \times 10^8}{0.04} = 7.5 \text{ GHz}$$

$$f_{11} = \frac{c}{2} \sqrt{\frac{1}{a^2} + \frac{1}{b^2}} = (1.5 \times 10^8) \sqrt{(1/0.03)^2 + (1/0.02)^2} = 9.01 \text{ GHz}$$

$$f_{20} = \frac{c}{a} = 10 \text{ GHz} \quad (> 9.5, \text{ not excited})$$

**Modes excited (carrier  $\hat{f}_c$ ):** TE<sub>10</sub>, TE<sub>01</sub>, TE<sub>11</sub>, TM<sub>11</sub>. TE<sub>20</sub> is just above cutoff so it does not propagate.

**Group velocities:**  $u_g = c\sqrt{1 - (f_{mn}/f)^2}$  with  $f = 9.5$  GHz.

| Mode                                | $f_c$ (GHz) | $u_g/c$                          | $u_g$ (m/s)        |
|-------------------------------------|-------------|----------------------------------|--------------------|
| TE <sub>10</sub>                    | 5.00        | $\sqrt{1 - (5/9.5)^2} = 0.85$    | $2.55 \times 10^8$ |
| TE <sub>01</sub>                    | 7.50        | $\sqrt{1 - (7.5/9.5)^2} = 0.61$  | $1.84 \times 10^8$ |
| TE <sub>11</sub> , TM <sub>11</sub> | 9.01        | $\sqrt{1 - (9.01/9.5)^2} = 0.32$ | $0.95 \times 10^8$ |

**Travel times:**  $t = L/u_g$  over 100 m.

| Mode                                | $t$ ( $\mu$ s) |
|-------------------------------------|----------------|
| TE <sub>10</sub>                    | 0.39           |
| TE <sub>01</sub>                    | 0.54           |
| TE <sub>11</sub> , TM <sub>11</sub> | 1.05           |

Higher modes arrive later because they propagate closer to cutoff, so  $u_g$  drops.